International Journal of Data Science and Analytics
https://doi.org/10.1007/s41060-021-00272-2

APPLICATIONS

An Information-theoretic approach to dimensionality reduction in data science

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Received: 26 December 2020 / Accepted: 29 May 2021

Abstract

Data reduction is crucial in order to turn large datasets into information, the major purpose of data science. The classic and richer area of dimensionality reduction (DR) has traditionally been based on feature extraction by combining primary features in a linear fashion, aiming to preserve or maintain covariance/correlations between the features. Nonlinear alternatives have been developed, including information-theoretic approaches using mutual information as well and conditional entropy based on target features. Here, we further this approach to feature selection or reduction strategy based on the concept of conditional Shannon entropy of two random variables. Novel results include (a) a dimensionality reduction method based on conditional entropy between predictors themselves along two variants, disregarding the influence of the target feature; (b) an error-prevention method inspired by error-detection and correction in information theory for DR with genomic data that can be used for abiotic data as well; and (c) a comparative assessment of the performance of several machine learning models on input features selected by these methods. We assess the quality of the techniques based on their performance in solving three application problems (Malware Classification, BioTaxonomy, and Noisy Classification) of various degrees of difficulty with competitive outcomes. Some useful heuristics arise from the analysis of the results and also suggest some problems of interest for further research.

Keywords Dimensionality reduction · Feature selection · Shannon entropy · Noncrosshybridizing DNA bases · Malware classification · Biological species identification

1 Introduction

Our ability to generate, gather and store volumes of data across many fields has far outpaced our ability to derive useful information from the data with current computational resources. Often times, the available data include a large number of features, not all of which are necessarily needed or even helpful to solve a given problem. For example, two features may be equivalent, one may be redundant (e.g., contain information derivable from others), and some may even be wholly irrelevant to the solution(s). Therefore, data reduction is crucial in order to turn large datasets into information, the major purpose of data science. The goal of dimensionality reduction (DR) is to identify and select or derive most useful features in a given corpus of data for a given problem concerning a population based on available and present data (usually, in a relatively small sample.) This goal is difficult to achieve in abstract since a number of different problems can be posed on exactly the same data, possibly with contradictory aims. We present here an information-theoretic approach to DR as a feature selection strategy based on the concept of conditional Shannon entropy of a random variable (RV). We are only concerned with exploring the effectiveness of these

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entropy methods rather than selecting features for an optimal solution to a given problem regardless of the criteria used.

The problem of reliable telecommunication led to several issues concerning a noisy channel. Shannon proposed a concept of *information* and solutions in his foundational paper [48], which gave rise to the field of Information theory. This theory is based on a critical concept called Shannon entropy that quantifies the degree of uncertainty of a random process. This metric provides the mathematical foundations for information theoretic analyses of channel capacity that characterizes the maximum amount of information being transmitted through a noisy channel while allowing noise removal without loss of information [48]. It has been also interpreted as a measure of the degree of *randomness* and/or *diversity* in a stochastic process. The concept of entropy itself can be traced back to [9], later used in physics for heat theory and thermodynamics by [3], but here it will only refer to Shannon entropy and be denoted H.

Entropy has been applied in many other fields including digital communication, finance, computer science, biology and many other fields. [40] initiated a systematic application of this concept for portfolio selection in finance. Later on, [34] replaced arithmetic mean with geometric mean to measure return in assessing a portfolio and introduced the concept of *incremental entropy* for portfolio selection as a more objective and testable criterion. Similarly, [60] explored portfolio selection problems leveraging the hybrid entropy for asset risk estimation. In finance, this concept is now crucial to portfolio selection and asset pricing [65].

Likewise, the field of computational biology and bioinformatics uses information theory to answer some important fundamental questions in genomics, transcriptomics, proteomics and metabolomics [6] given that the degree of uncertainty in biological processes is pervasive due, in particular, to DNA mutations in living organisms [52]. In particular, the concept of speciation is core in evolutionary biology and causes a particular group of organisms to diverge from their ancestors in such a way that entirely new species may eventually arise [49]. However, reconstructing the evolutionary history of new species is very difficult. [50] deployed an information theoretical framework to analyze maternally inherited haploid and bi-parentally inherited diploid nuclear markers and extended that analysis to construct minimum-spanning networks for both genomes. Similarly, [7,27,56] presented global sequence analysis methods inspired by information theoretic frameworks with natural language underpinnings (such as block-entropy or L-tuple entropy), chaotic game representation of genomic sequences [55] and local sequence analysis techniques such as motif classification and prediction of transcription factor binding sites.

Another recent trend in bioinformatics has led to the application of these frameworks to analyze genomic sequences based on alignment-free methods, for instance by [20], who presented a novel alignment-free approach to classify genomes based on measurement of the similarity among genetic sequences using elements from both word-rank order-frequency statistics and information theory. More recently, [19] presented a novel alignment-free method to extract significant information from DNA sequences of organisms based on the hybridization affinity of these sequences to a judiciously selected set of oligonucleotides of the same length (more details below.) The same conceptual framework was used to answer some fundamental questions in biology involving phylogenetic reconstruction [18], phenotype prediction [28], environmental feature encoding in genomes [30] and species identification [29]. The answers to these questions demonstrated that minimization of Shannon entropy is a very useful criterion to select appropriate biomarkers to capture signals from DNA sequence by hybridization affinity that allows higher quality solutions to these problems with substantially fewer features.

Entropy has also found wider applications in the field of Data Science, in particular for feature selection to reduce data to a set of significant and manageable features. The purpose here is to improve the generalization ability of a machine learning model so that it could be more robust to new data and explainable to its users. The approach has primarily used the concept of *mutual information* $I(Y, X)$ (the residual uncertainty left in a random variable Y conditioned on information about the variable X) to quantify information independence between two variables, analogous to the concept of statistical independence. This notion of independence has two further advantages, i.e., it captures nonlinear relationships between the variables and is invariant under transformations in the feature space [54]. Using this approach, [25] proposed a new method to calculate mutual information between input and class variables based on the Parzen window to select significant features for classification problems. Several multilayer perceptrons trained with the conventional back-propagation learning algorithm with various number of features ranging from 2 to 12 were trained and assessed on multiple datasets (about Sonar, Vehicle, Letter, Breast Cancer, Waveform and Glass) with at least 68% accuracy. Similarly, [5] proposed a unifying framework to select features based on information-theoretic concepts by viewing the problem of selecting significant features as a conditional likelihood of a target feature on the given features. Also, [13] proposed a model for dimensionality reduction method using a 'supervised algorithm' to find a projected space maximizing the mutual information between predictors and a target feature. Likewise, [46] proposed a method to extract features based on a componentwise gradient ascent method using mutual information between predictors and a target feature. Furthermore, [4] proposed a novel algorithm for dimensionality reduction based on quadratic nonparametric mutual information between predictors and the corresponding class labels in the graph embedding framework. Likewise, [51] proposed another feature selection method based on mutual information criterion to train machine learning models to improve the prediction accuracy and shorten the training time. Despite differences, these models are solely based on mutual information between predictors and a target feature.

In this paper, we explore two variants of the same idea. The classic and richer area of DR has traditionally been based on feature extraction by combining primary features in a linear fashion, aiming to preserve or maintain covariance/correlations between the features, as can be found in a variety of sources, e.g. [22,23,31]. Here, we are more interested in the rather unexplored area of feature selection from the perspective of information theory. New contributions are

(a) a dimensionality reduction method based on conditional entropy *between predictors themselves* along two variants, disregarding the influence of the target feature;
(b) an error-prevention method inspired by error-detection and correction in information theory for DR with genomic data that can be used for abiotic data as well;
(c) a comparative assessment of the performance of several machine learning methods on these methods.

In Sect. 2, we present two different applications of entropy and assess them on representative problems in order to get a more complete picture of this information-theoretic approach for DR. Next, we describe three representative problems from various domains and degrees of difficulty and suitable datasets to be used for solutions of these problems. They include Malware classification (MC), BioTaxonomy (BT) and Noisy Classification (NC), defined precisely below in Sect. 2. They will be used to benchmark their effectiveness in Sect. 3. Finally, we compare and contrast these solutions in Sect. 4 and draw a number of heuristic conclusions on the overall effectiveness of the DR methods for similar kinds of problems, including a ranking by their overall effectiveness across problems.

2 Information-theoretic methods

The motivation behind our approach to nonlinear dimensionality reduction is twofold. Intuitively, a quantification of how informative a feature is could be used to perform feature selection better. Theoretically, this idea makes no sense unless an objective definition of *information* content is provided. Shannon provides an intriguing way to solve this problem, using the concept of conditional (Shannon) entropy of two random variables (RVs). The information content $I(a)$ provided by an observation $[X = a]$ of a RV X is $I(a) = -log_2(p[X = a])$, where $p[X = a]$ is the probability of the event associated with an observation of a value a for the RV X. The (Shannon) *entropy* H of a RV X is the expected value of the information content of the observations of all values of X and can be regarded as a measure of the uncertainty in determining any given outcome of an observation of X, or its quantification as the average number of bits necessary to identify all possible (unique) values of X (somewhere between 2 and 3 bits for the 6 outcomes of the roll of a die. If natural logarithms are used, the unit is called "nats" not bits and this unit will be used in reporting entropies below.) The *conditional entropy* $H(Z \mid X)$ of a RV Z on a given RV X is the average entropy of Z conditioned on the observation of a value of X, i.e., the expected value $E(H(Z_a))$ of the entropies of the RVs given by $Z_a : [Z \mid X = a]$ across all possible values of a in the range of X. This entropy can be interpreted as the *uncertainty left in the values of* Z *given the observations of co-variate feature* X in a given data point Z_a, averaged across all data points. Conditional entropy is thus not a symmetric function of the RVs involved and allows selection of one feature over another. A potential disadvantage will be offset by considering both $H(X_i \mid X_j)$ and $H(X_j \mid X_i)$. The conditional entropy can thus be generalized to any number of conditions, i.e., $H(Z \mid X_1, X_2, \ldots, X_m)$ can be interpreted as the average uncertainty left in feature Z given the values of joint observations $X_1 = x_1, X_2 = x_2, \ldots, X_m = x_m$ in the same data points. The R package *infotheo* was used to compute Shannon entropies below using the Miller–Madow asymptotic bias corrected empirical estimator (which requires discretized features) and using the discretization option *equalfreq*, where necessary.

Although these definitions of (conditional) entropy can be extended to continuous-valued RVs, we will restrict ourselves to the discrete case for three reasons. First and foremost, data science is currently mostly about data being processed on digital computers, so data must be discretized even if it is continuous at the source. Second, most data are captured by digital sensors and transmitted via digital networks, so it will be discretized in the process anyway. Third, the classic assumptions for data analytics (e.g., normality) are not necessary for discrete data and the more common machine learning solutions available today.

2.1 Reduction by conditional entropy

In this reduction, informational independence between features is quantified using Shannon conditional entropy $H(X_1 \mid X_2)$ between features X_1, X_2. When this entropy is high, knowledge of the values of X_2 remove little of the uncertainty in the values of X_1 and selection of X_1 may be necessary even if X_2 has been included. Conversely, when this entropy is zero (or close to), X_2 (essentially, respectively) determines X_1 and thus X_2 is an informative feature and would suffice because models could derive any information in X_1 from it. This idea can be implemented in various ways. (Examples are given below after introducing some data sets for evaluation.)

2.1.1 Reduction by single (H + m) and paired conditional entropy (H + m +)

We are interested in ascertaining target feature values given some reduced feature list from the full list of predictors $X_1, X_2, \ldots, X_m$ in the given data. To use as few Xs as possible, we are interested in selecting only those predictors that are as informative as possible. In particular, they should be as independent from other predictors as possible. Information-theoretically that means that the conditional entropies relative to the other predictors (i.e., the uncertainty left in the predictor given another predictor's value) should be high overall. Thus, to decide whether to include X_i, we computed the average

$$avg_i = avg \ \ \{H(X_i \mid X_j) : 1 \leq j \neq i \leq m\}$$

of the $H(X_i \mid X_j)$ over all other X_j to quantify how informative predictor X_i is compared to others, for each i in the range of m predictors (excluding the target feature.) For a given dataset, we can thus select the top features *maximizing* this average to select as many features as desired, after sorting the list in decreasing order. A second *paired* variant on the same idea to quantify the information content of X_i is to compare the double conditional entropies $H(X_i \mid X_j, X_k)$ overall j and k instead. If the average of these entropies is high, X_i should be selected, as above. We can repeat the same process to select more features as needed.

2.1.2 Reduction by iterated conditional entropy (H + m ++)

Another interesting variation is to select features based on the following recursive procedure.

1. Select the first feature X_{i_1} indexed by i_1 as before to maximize the average single conditional entropy: $i_1 = arg_{1 \leq i \leq m} \ Max_i\{avg_j \ H(X_i \mid X_j)\}_{1 \leq j \leq m}$;
2. Having selected $X_{i_1}, \cdots, X_{i_k}$ ($k \geq 1$), select the next feature $X_{i_{k+1}}$ indexed by i_{k+1}:

$$i_{k+1} = arg_{i \in [1:m]-\{i_1,\cdots,i_k\}} \\ Max_i\{avg_j\{H(X_i \mid X_{i_1}, \cdots, X_{i_k}, X_j)\}_{1 \leq j \leq m}\}.$$

This procedure can be repeated to select additional features as needed.

2.2 Reduction by conditional entropy for the target feature

In statistics, when selecting features for predictors, the standard approach is to take the target feature into account. Here, we follow a similar method in order to make a fair comparison of the performance of our new approach to those methods. In a variant of our previous method, the degree with which a target Y is predictable given a feature X_i is decided using Shannon conditional entropy $H(Y \mid X_i)$ between Y and feature X_i. When this entropy is low (high), the uncertainty is very low (high, respectively) for the values of Y given the values of X_i, so X_i is very informative (or is not, respectively.) Hence, X_i should now be selected only if this entropy is low. We implemented this approach in the similar way as in the case of the reduction by conditional entropy above, with the following changes:

1. Calculate the $H(Y \mid X_i)$ overall X_i to quantify how informative predictor X_i is compared to others;
2. Select the top features *minimizing* this entropy after sorting the list in increasing order.

We used similar notation (without the m) to represent these counterpart variants excluding target, i.e., Single/Paired conditional entropy including target by $H/H+$ and Iterated conditional entropy by $H{++}$.

2.3 Reduction by noncrosshybridizing bases

A virtually inexhaustible source of data lies in the biological diversity of life on earth, particularly large genomic data that holds the blueprint of most living organisms on earth. Microarrays are regarded as a powerful tool to capture and mine large-scale genomic data. They are planar substrates such as glass, mica, plastic or silicon, where DNA strands are affixed to allow specific bindings of biosamples collected from an organism and undesirable interactions between them (see [45] for more detail.) The information gathered by these tools has wide applications in the fields of biology, medicine, health and scientific research. (Although biologists refer to probes to indicate the biosample and target to indicate the DNA sequence fixed on the microarray, here the terms probe and target are used with the reverse meaning, i.e., the probes are fixed to the DNA chip.)

Despite the advantages offered by microarrays, the analysis relying on these data gives results that are hardly reproducible because of the likely high uncertainty of hybridization of targets to probes. No constraints are usually implemented in these chips to minimize unintended (cross)hybridization between probes and/or targets. Consequently, the results become unreproducible and hence unreliable, as argued in [19]. A second disadvantage of cur-

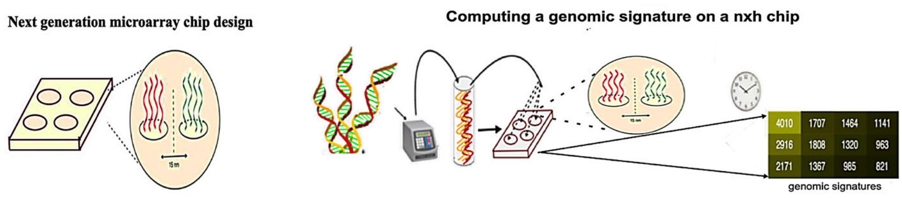

Fig. 1 Nxh DNA chip design (reproduced with permission from [28]). (Right) Workflow to obtain a genomic signature of a DNA sequence using an nxh chip design containing four probes (arranged as shown inside the oval), as described in [18,28]

Table 1 Nxh bases for genomic coordinate systems

Basis	Length	Probes	τ	Entropy
3mE4b	3	4	1.1	0.45
4mP3-3	4	3	2.1	0
8mP10	8	10	4.1	0.57

rent microarray designs is that they might miss target strands if they do not hybridize to any probe, and thus miss signals that could yield useful information; for example, if probes are Watson–Crick (WC) palindromic, they are more likely to hybridize with themselves.

[19] and [17] have addressed these drawbacks of conventional microarrays, using other ideas from information theory, namely error-detecting and error-correcting codes. DNA sequences diffuse in solution (the channel) "looking" for a (WC) complementary sequence to hybridize to. If the probes attached to the chip are not carefully selected, we get the equivalent of a channel allowing errors in the intended hybridizations when a target encounters the wrong probe first (i.e., the hybridization affinity to the probe is not optimal.) In order to find the equivalent of error-detecting/correcting codes, we need a metric for hybridization affinity. However, the standard Gibbs energy of duplex formation (e.g., as computed by the Nearest-Neighbor model in [44]) does not have the properties of a metric, not to mention that the Hamming distance for DNA would require alignments of the sequences, an assumption utterly inadequate for DNA in solution where frameshifts prevail. [15] and [39] introduced an appropriate metric, the so-called *h-distance*. To have the distance properties, oligonucleotides (k-mers) must be paired to their WC-complements as single units (so-called *pmers*), otherwise different oligos would be at distance 0. The problem of finding large sets of DNA hybridization independent (or orthogonal) oligonucleotides is thus analogous to finding error-*preventing* codes and was formalized as the CODEWORD DESIGN problem in [16,17], who proved it to be very hard (technically, **NP**-complete to solve in full generality—see [39]).
A solution is called a *noncrosshybridizing (nxh) basis*, and they would be extremely useful in the design of next generation microarrays. A nxh basis [design shown in Fig. 1(Left)] can be defined as a set of pmers minimally covering the entire DNA space of a given size n for hybridization events. More precisely, a set of pmers (referred to as *probes* below) is an nxh basis if it consists of (a) a sufficient number of pmers in such a way that every random n-mer of the same length n in any target will hybridize with at least one (of the oligos in a probe (consisting of multiple copies of a basis oligo or of its complement on the chip); and (b) any random n-mer in any target will hybridize with at most one oligo in a probe, under appropriate reaction conditions (as determined by a stringency threshold τ.)

Despite the difficulties to meet these requirements, a handful of nxh bases have been found for short oligos of size $n \leq 8$ based on h-distance by [15] as a model of hybridization between any pair of pmers. The bases used with this DR method here are shown in Table 1.

With an nxh basis design in hand, a DNA sequence x can be first shredded to fragments of a uniform length n (same as that of the probes.) Then, these fragments are tested for hybridization with the probes and a readout can be obtained as a vector of dimension equal to the number of probes in the basis. A feature value in the readout is given by the number of shreds hybridizing with the corresponding probe in the basis. Finally, the readout is normalized with the partition function so that all values add up to 1. This normalized vector is the reduction in the original DNA sequence x and is referred to as the *genomic signature* of x. Genomic signatures are thus a set of derived features extracted from DNA sequences by hybridization of their fragments to a set of pmers selected *a priori*. The process of computing such a signature is shown in Fig. 1(Right). The nxh bases can then be regarded as a sort of dimensionality reductors for DNA data. How informative are they?

The quality of these nxh bases can be quantified using the entropy of the corresponding distribution (i.e., the degree of uncertainty with which fragments attach to a single probe.) Ideally, we would prefer a nxh chip design (such as 4mP3at2.1 in Fig. 2 below) where the entropy is 0 since each fragment attaches to exactly one probe with certainty. The more uncertainty (noise), the larger the entropy and the lesser the reliability in reproducing the same signature (due to the inherent randomness in the diffusion process of DNA in a solvent, such as water.) The entropies of these bases were computed *in silico* on an HPC. This method affords a solution to the problem of high entropy posed by the use of the full set of k-mers, as described in [8], which leads to low accuracy in the species classification problem.

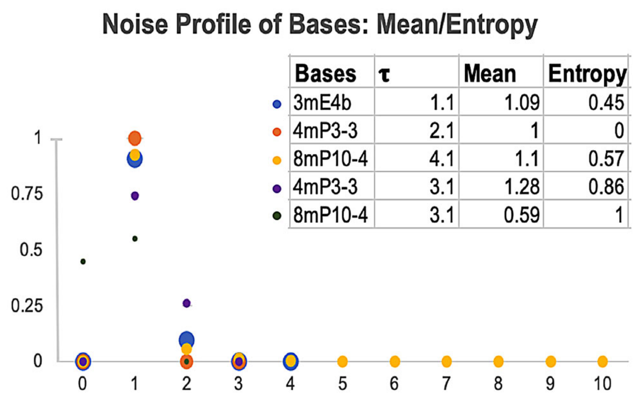

Fig. 2 Reliability analysis of nxh chip based on two metrics, namely the expected value and Shannon Entropy of the random variable N that counts the number of probes in a basis (x-axis) that a pmer selected at random could stick to any probe (shown in the y-axis as a fraction of all possible pmers), for five nxh bases, as given in [18]. The sample space consists of all the possible pmers of the same length as that of the probes. Ideally, this random variable should be constant of value 1 (no ambiguity in the hybridization process for a noise-free chip.) The Shannon Entropy quantifies the degree of uncertainty (noise) with which a target shred sticks to a single probe. This value should equal 0 for an ideal chip design

In order to quantify the quality of this chip, consider the following random experiment to be performed multiple times. The sample space for the experiment contains all possible pmers of length that of the probes in the basis. The experiment selects a pmer from the sample space at random, with a uniform distribution. The random variable N used in this quantification observes the total number of probes that a random pmer sticks to under the hybridization model, where two pmers hybridize if and only if their h-distance is less than a certain threshold τ. The quality of any basis can be quantified by the entropy $H(N)$. Ideally, a basis would show a random variable N with values constantly (or at least its expected value is) equal to 1, and its Shannon Entropy is thus equal to 0. The quality profile of several bases is shown in Fig. 2.

2.4 Assessment methods and data sets

We describe next three problems that will be used to assess the quality of the DR methods just described.

2.4.1 Malware classification

We evaluate our DR techniques first with the task of malware classification. Specifically, we use the Microsoft Malware Classification Challenge [43] dataset that was released in 2015 and is publicly available through Kaggle. The dataset consists of a set of 10,868 known malware files representing a mix of 9 different malware families as summarized in Table 2. Each datapoint contains both the raw binary content of the malware files as well as metadata information extracted using the IDA disassembler tool for a total of 1804 raw features. The classification challenge is to correctly assign every piece of malware to one of the 9 possible malware types defined with the challenge, shown in Table 2. We used different types of predictors, as described above in Sect. 2. Designing a classifier with the full set of features is very difficult.

Table 2 Summary of the dataset for Microsoft's Malware Classification Challenge

Malware class	Size	Type of malware
Ramnit	1541	Worm
Lollipop	2478	Adware
Kelihos_ver3	2942	Backdoor
Vundo	475	Trojan
Simda	42	Backdoor
Tracur	751	TrojanDownloader
Kelihos_ver1	398	Backdoor
Obfuscator.ACY	1228	Any kind of obfuscated malware
Gatak	1013	Backdoor

Selection by iterated conditional entropy was repeated twelve times to obtain a selection of 12 features, so that $X_{i_{12}}$ maximizes $H(\,X_{i_{12}}\,|\,avg_i\,X_{i_1},\ldots,X_{i_{11}}\,)$ among the $m - 11 = 333$ remaining features, after excluding the first 11 selections of $X_{i_1},\ldots,X_{i_{11}}$.

2.4.2 BioTaxonomy

In biology, classifying living organisms into species is a fundamental problem, called biological taxonomy (BioTaxonomy), first addressed by [26]. Linnaeus introduced the idea of cataloging life on earth into a hierarchical classification system. Regardless of the differences in granularity, the concept of *species* underlines them all and is the most fundamental [59]. However, the concept of species is so diverse that it has proven to be extremely difficult to define. For example, more than 25 definitions were proposed in the past 25 years alone by [11,24,35,37,57] and [59], particularly because such a definition depends not only on an organism's genome, but also on its morphology, behavior, geographic distribution, ecology and perhaps even natural history [10,12,14,41,57]. The current practice in biology for such a definition is to act by consensus on a group of organisms (a taxon) at a time and organize them together into what is referred to as the "standard biological taxonomy." Thus, we will assume that the species of an organism is given by its label on GenBank [33] as the ground truth.

Standard taxonomies today also consider metagenomic factors, (such as geography and climate) and use as major taxa *genus, family, order, class, phylum, kingdom* and

Table 3 Summary of organisms in the sample data for BT

SpecID	Species	No. of specimens
1	*Apis mellifera*	4
2	*Arabidopsis thaliana*	5
3	*Bacillus subtilis*	18
4	*Branchiostoma floridae*	18
5	*Caenorhabditis elegans*	6
6	*Canis lupus*	18
7	*Cavia porcellus*	4
8	*Danio rerio*	9
9	*Drosophila melanogaster*	18
10	*Gallus gallus*	18
11	*Heterocephalus glaber*	3
12	*Homo sapiens*	18
13	*Macaca mulatta*	8
14	*Mus musculus*	18
15	*Neurospora crassa*	5
16	*Oryza sativa*	12
17	*Pseudomonas fluorescens*	6
18	*Rattus norvegicus*	18
19	*Rickettsia rickettsii*	18
20	*Saccharomyces cerevisiae*	18
21	*Zea mays*	7

domain. However, recent research in [29] and [30] has pointed out the fact that information stored in DNA contains significant clues about phenotypic features and also environmental factors responsible for determining morphological features beyond the genealogical lineage of a living organism. Therefore, it is reasonable to expect that there might be sufficient information in the genome of an organism to identify the species it belongs to, as suggested by [28]. Thus, to evaluate DR methods, the challenging problem of species identification, i.e., ascertaining the species of an organism, given only by a representative DNA sequence in its genome only, will also be examined.

For a dataset, representative genes were obtained from mitochondrial DNA (specifically, cytochrome Oxidase genes COI, COII and COIII, CytB) of 249 organisms collected from the GenBank. These organisms are distributed across 21 species as shown in Table 3. We used different types of predictors using pmer counts and nxh bases, as described above.

2.4.3 Problem: classification on noisy data

This classification problem was designed with a synthetic data set that allows us to have perfect prior knowledge of *feature dependencies* that will remain hidden from the DR methods. Their effectiveness will then be assessed by how well they discover these hidden but most relevant and independent features from a full set that includes other confounding features derived from a few primitive ones.

The synthetic data sets were designed to facilitate this assessment using the method described in [21] and publicly available in python at
https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_classification.html.
The first dataset consists of 1000 points, 22 predictors X_1 , . . ., X_{22}, and one dependent categorical variable, *Label*. The second data set was generated likewise but halving the number of all parameters involved in generating features in the first set, so we just describe the larger set. Each dataset was generated in two phases. First, we used the sklearn API to generate 12 primitive features in a 12D hypercube with sides of length $1.6 = 2 * 0.8$ and assign about the same number of points to each of the eight classes. Each class is composed of several Gaussian clusters located near corners of the 12D hypercube. For each class, the informative features are drawn independently from the standard normal distribution $N(0, 1)$. Four more features X_{13}, X_{14}, X_{15} and X_{16} were generated as linear combinations (with random coefficients) of some primitive features, as provided in the API. In the second phase, we generated 6 more predictors as follows. Features X_{17} and X_{18} were drawn as a repeat of two randomly selected primitive features/columns. X_{19} was generated as the sum of squares of two features selected randomly as

$$X_{19} = X_i^2 + X_j^2 \quad \text{for some} \quad \text{fixed} \;\; 0 \le i, j \le 12 .$$

X_{20} was obtained from the values of a linear regression model fitted using two randomly selected features X_i and X_j as predictors with the *Label* as response. The value for X_{20} was the deviation between the pair of values in *Label* and the regression prediction. X_{21} was obtained the same way but with X_i^2 and X_j as predictors from another random selection of predictors X_i and X_j. Finally, X_{22} was generated as the outcome of the natural *logarithm* of a randomly selected but uniform predictor X_i (a transformation that does not change entropy.) Again, we used different types of predictors as described above.

2.5 Encoding of data

2.5.1 Encoding of malware to DNA

A piece of malware can be regarded as an ordered set of hexcodes representing a string of hexadecimal characters. Thus, a DNA encoding can be simply obtained by converting each hexadecimal character to its binary equivalent, and then concatenating these binaries into a sequence in DNA form. Two bits encode four possible strings that can be regarded as

Table 4 Solutions used to evaluate DR methods based on their average performance (F1-Score/Precision/Recall) on 12 features selected from 344 features (described above in the Assessment Methods section)

Solution models	Software	Package/Module
Linear Discriminant Analysis (LDA)	R	MASS [53]
Support Vector Machine (SVM)	R	e1071 [32]
Multinomial Logistic Regression (MLR)	R	nnet [42]
Gaussian Naive-Bayes (GNB)	Python	sklearn [36]
Random Forest (RF)	Python	sklearn [36]
k-Nearest Neighbors (kNN)	Python	sklearn [36]
Neural Networks (NN)	Python	H2O [2]

a,c,g, or t, i.e.,

$$00 \rightarrow a;\ \ 01 \rightarrow c;\ \ 10 \rightarrow g;\ \ 11 \rightarrow \ t.$$

2.5.2 Encoding of DNA to feature vectors

To solve the BioTaxonomy problem, we shredded each DNA sequence into fragments of length 3. Each of these fragments represents a codon responsible for synthesizing a unique protein. We paired each codon with its WC complement forming 32 3-pmers (32 3-mers paired with their WC-complements) and then counted the frequency of these 3-pmers in the sequence. Each vector containing such counts was then normalized using the partition function to make the sum of all features equal to 1. These normalized vectors were then used for DR using the entropy-based reduction methods discussed above.

3 Assessments and results

In order to assess the quality of the DR methods described in Sect. 2, we applied them to the data sets for the three problems described above and show the results in this Section. As is typical in machine learning, we split the data sets into a training and testing subsets in proportion 80–20%. Each of the seven solutions in Table 4 was run using fivefold cross-validation to fit a model for each kind of solution. (We are assuming a prior knowledge of these machine learning solutions used below, see [47] for details.) Performance for the solutions obtained was scored using accuracy, precision, recall and weighted F1 scores, with some adjustments, i.e., we used the proportions of data in each class to the entire data set (partition function) as weights and the procedure was repeated 100 times for each score.

The results are shown in Figs. 3, 4, 5 and 6 and are discussed next.

3.1 Results: Malware classification

On the problem of MC, DR methods using single conditional entropy (H+m, no target feature), iterated conditional entropy (H++, including target) and iterated conditional entropy (H+m++, no target) performed consistently better (F1-score over 80%, except for LDA and MLR) than others across all solutions methods in Table 4. DNA method 3mE4b also performed consistently well except with neural network models (NN), although 8mP10 surprisingly did not despite having more and longer probes. This is true regardless of whether we are reducing to 6 or 12 features. These scores are far better than a control we run on random selections of 6 and 12 features from the full set of 344 (F1 scores under 20%.)

In terms of solution models, the top performers were RF (F1-scores over 90%, unsurprisingly), kNNs (F1-scores over 85%, except on H+m+ and 8mP10) and SVMs (F1-score over 75%, except for 8mP10.) Models LDA and GNB performed very inconsistently (F1-scores between 46–98% and 10–90%, respectively.) MLRs and NNs fell in between, with a moderate performance.

3.2 Results: BioTaxonomy

On the problem of BT, all DR methods using conditional entropy have comparable high performance (F1-score over 80%, except with neural network models (NN), where DNA method 3ME4b performed better), although 8mP10 did not perform well either. Here, the performance on 6 reduced features was consistently lower than 12 features across all solutions methods.

In terms of solution models, the top performers were RF (F1-scores over 94%, except for 8mP10), kNNs (F1-scores over 90%, except on 8mP10) and MLRs (F1-scores over 90%, except for 8mP10.) Models SVM and NNs performed poorly (e.g., F1-scores below 45% on 8mP10 and between 42 and 67%, respectively.) Models LDA and GNB performed very inconsistently (F1-scores between 43–94% and 41–91%, respectively.)

3.3 Results: Noisy classification

On the problem of NC, the performance of DR methods using conditional entropy (H+, including target feature), iterated conditional entropy (H++, including target), paired and iterated conditional entropy variants (H+m+ and H+m++, no targets) was consistently good on both data sets SYN13 or SYN22 on 6 reduced features. For 12 features, all DR methods performed consistently and comparatively well (F1-score over 82%) consistently across all solutions methods for both data sets.

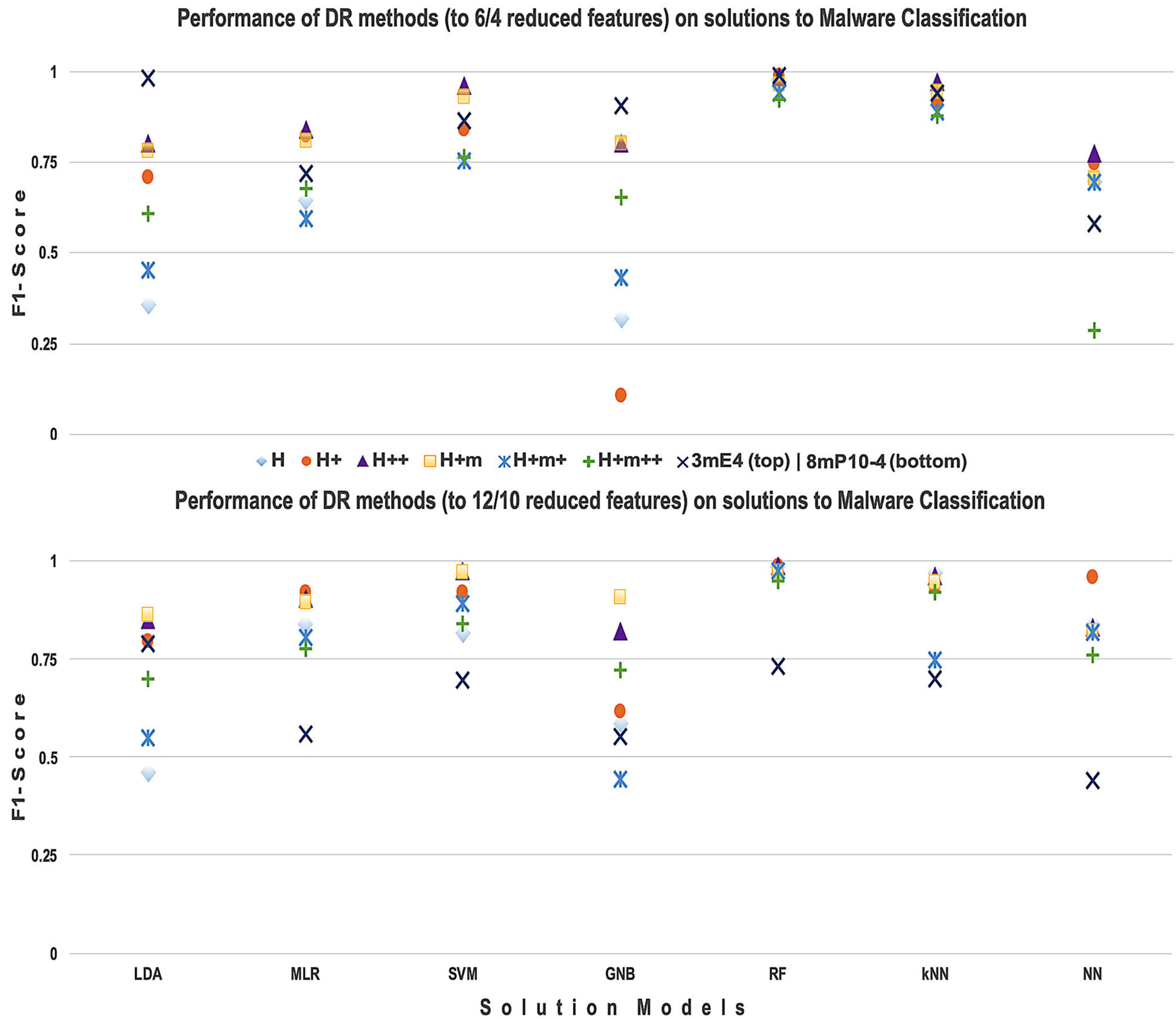

Fig. 3 Overall Performance of all DR methods based on conditional entropy for Malware Classification based on models trained on 6 features (top) and 12 Features (bottom) with genomic sequences for MS challenge data in Table 2. X-axis: solutions (LDA, MLR, SVM, RF, GNB, kNN, NN.) Y-axis: F1-scores. The eight DR methods are selection by conditional entropy on predictors only (H+m, paired H+m, iterated (H+m)++) or including the target features (H, paired H, H++, 3mE4b, 8mP10)

DR methods H+m, H+m+ and H+m++ discovered about 75%, 83%, and 85%, respectively, of the hidden primitive features in the raw data set with 12 primitive features (SYN22) for 12 reduced features. Similarly, DR methods H, H+ and H++ discovered about 75% of the hidden primitive features on the same dataset for the same number of reduced features.

Likewise, H+m, H+m+ and H+m++ discovered 50%, 100% and 100% of the hidden primitive features in the raw data set with 6 primitive features (SYN13) for 6 reduced features. However, the remaining methods could only capture about 50% of the primitive features on the same dataset for the same number of reduced features. Therefore, on average, DR methods H+m+ and H+m++ were found to be capable of capturing most of the primitive features while the remaining DR methods highly depend on datasets to do so. So, based on this criterion, it may be surprising that only methods H+m+ and H+m++ (*not using the target features*) did well.

Interestingly, the results on the dataset containing 12 reduced features from SYN13 were comparable to those on the dataset containing the same number of reduced features from SYN22. This demonstrates that, *occasionally noise present on the dataset might be helpful* for these models to solve a problem.

In terms of solution models, the top performers were RF (F1-scores between 75 and 99%), kNNs (F1-scores between 70 and 99%) and SVMs (F1-scores over 90%.) Models

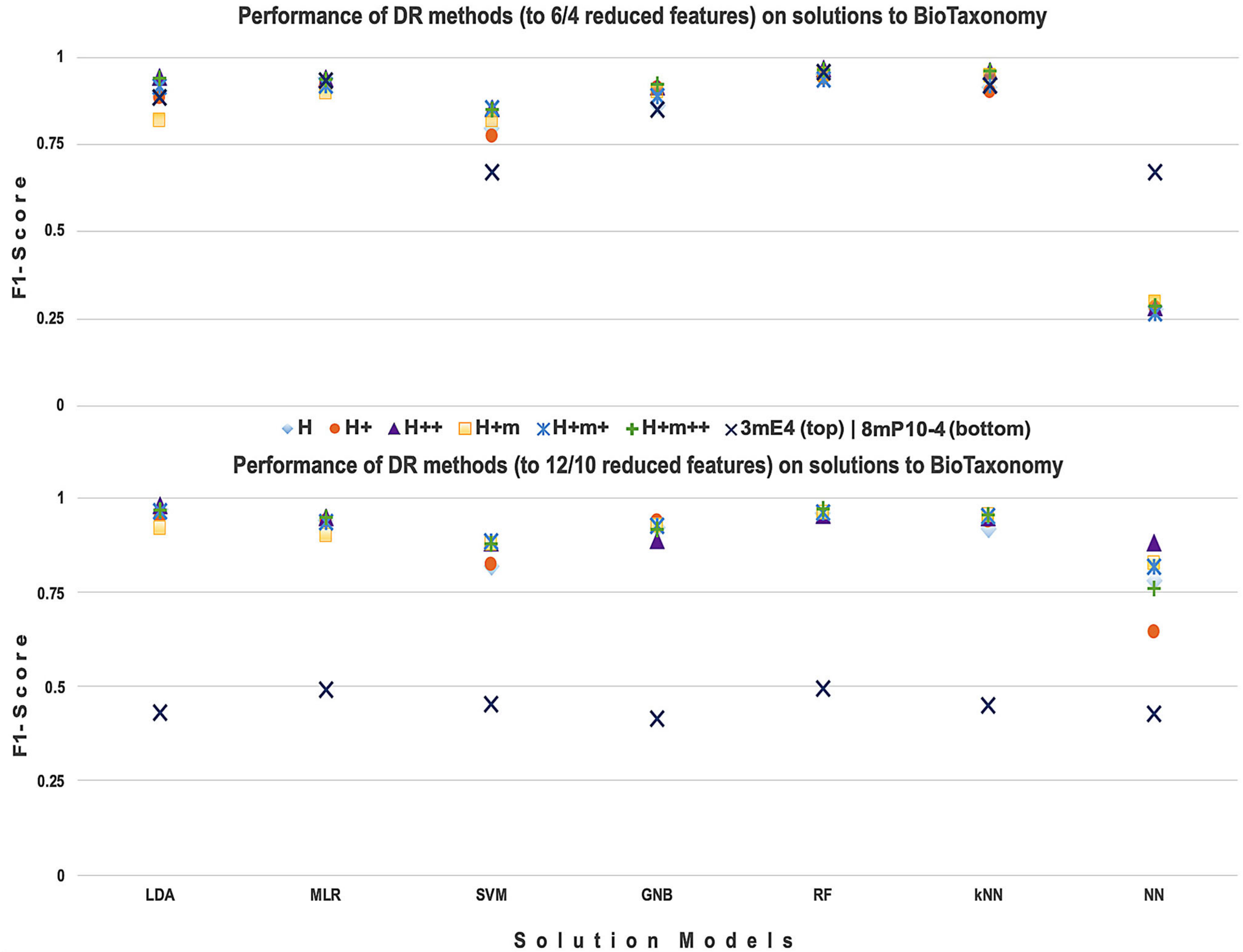

Fig. 4 Overall Performance of all DR methods based on conditional entropy for BioTaxonomy based on models trained on 6 features (top) and 12 Features (bottom) with genomic sequences for 21 species in Table 4. X-axis: solutions (LDA, MLR, SVM, RF, GNB, kNN, NN.) Y-axis: F1-scores. The eight DR methods are as shown in Fig. 3

MLR and LDA performed moderately well (e.g., F1-scores between 78 and 98% and between 40 and 100%, respectively.) Models GNB and NNs performed very inconsistently (F1-scores between 26–100% and 28–97%, respectively.)

4 Summary and conclusions

We have presented several novel variants of entropy methods for DR based on conditional entropy rather than mutual information, with three major applications for biotic and abiotic data analytics. All variants of informatic-theoretic DR presented here performed very well across the board compared to random selection of features. Perhaps surprisingly, conditional entropy of the target feature does not seem to provide any peculiar advantage over methods that only include predictors. Some nxh bases (DNA-based DR methods) produce comparable results for solving MC and BT problems. Therefore, we conclude that these DR methods are at least as effective as (if not more than) other conventional techniques for DR with much larger subsets of features. Moreover, these DR methods offer the additional advantage of being computationally efficient because they are parallelizable. To compute conditional entropies between a pair of features (both predictors or one predictor and another target), the information about other features is not required, hence several disjoint subsets of data can always be extracted and assigned to different computing nodes. This makes the process of feature extraction feasible, even for huge datasets.

We also presented a comparison of the solution models based on 6 and 12 features selected by entropy methods and four and 10 features extracted by DNA-based entropy methods. On average, solution models using only 6 features mostly performed poorly as compared to those using 12 features. Six (6) features might be too few for solving a complex problem, so reducing features too much may hurt for abiotic data. How-

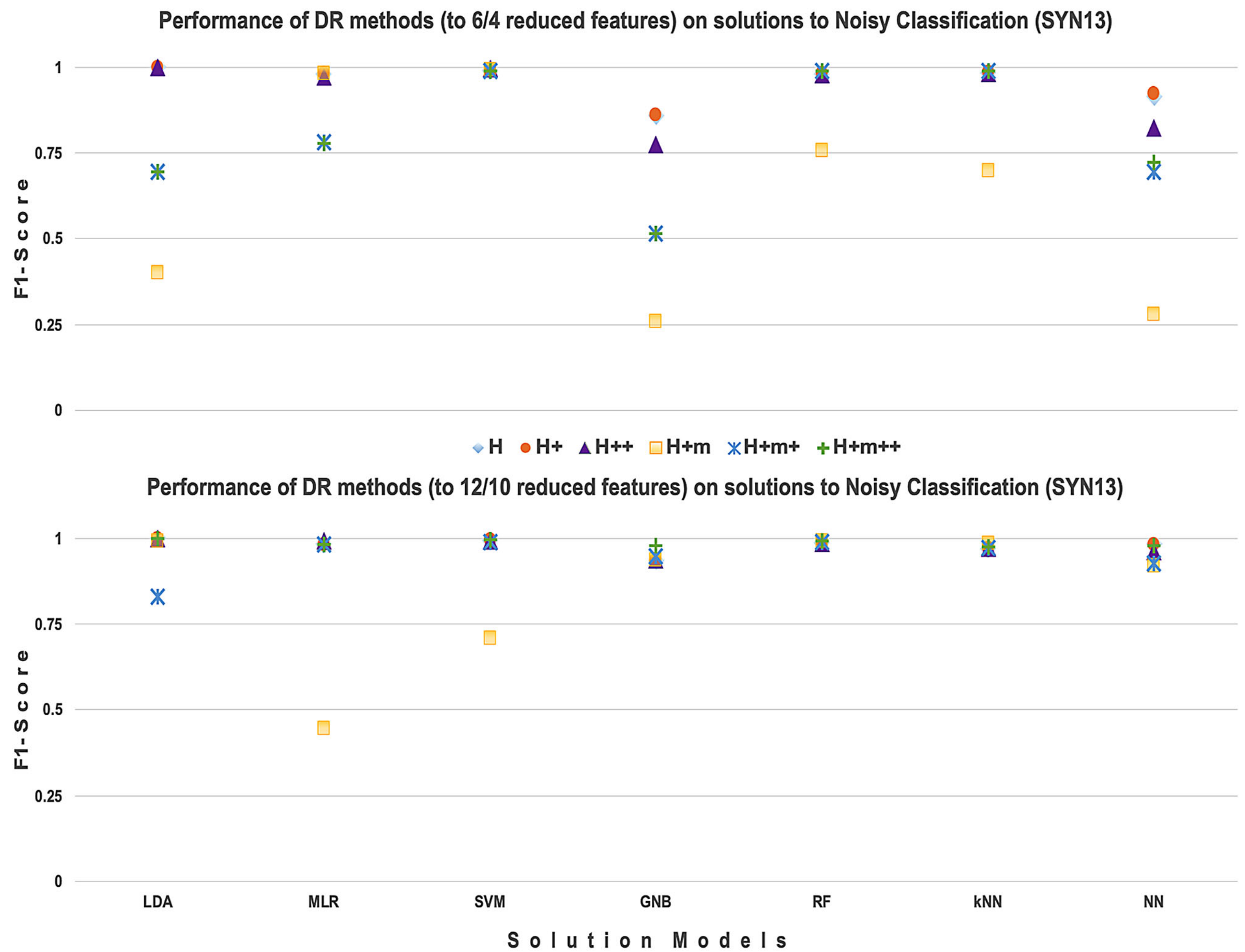

Fig. 5 Overall Performance of all DR methods based on conditional entropy for Noisy Classification based on models trained on 6 features (top) and 12 Features (bottom) with synthetic datasets, SYN13. X-axis: solutions (LDA, MLR, SVM, RF, GNB, kNN, NN.) Y- axis: F1-scores. The eight DR methods are as shown in Fig. 3

ever, we consistently observed that the performance scores for the solution models based on only 4 nxh DNA features were noticeably higher than those based on 10 nxh features. Therefore, we can safely conclude that the h-distance captures enough deep structure of DNA spaces that helps nxh DR extract really informative features comparable to that obtained by entropic methods on conventional data. These results make evident that *the sheer number of reduced features may not be as important as the impact of the structure of the data* in the quality of the solution to a problem.

Further, a comparison between the results obtained using variants of DR in this paper on both datasets for MC and BT problems allows us to conclude that DNA-based methods are just as effective as other entropy methods (if not better.) We were particularly careful not to perform any major data analysis while encoding the data into DNA (the encoding were straightforward) to ensure that any quality in the results is only due to entropy. It is possible that smarter encodings may yet further improve the quality of the DNA methods. In terms of solutions, some models (RFs, kNNs) perform better regardless of the problem (MC, BT or NC), while others (SVMs, LDAs and MLRs) perform very inconsistently across the three representative problems here.

There are other major advantages of these DR methods. First, for each data set, the best performing of these models are competitive with start-of-the-art for these problems (cited in the corresponding tables in the Appendix.) Second, the *running times on an HPC* of the relative entropy calculations to select features are in the order of minutes (single entropies), under 2 hours (paired entropies) and under 6 hours (iterated entropies); for nxh basis, feature extraction on an HPC (using a single node) also takes in the order of minutes for all data sets. Further, these times essentially scale linearly with the size of the data, except for iterated entropy, although the systematic good performance of iterated entropy methods (H++ and H+m++) comes at a dear price because the run-

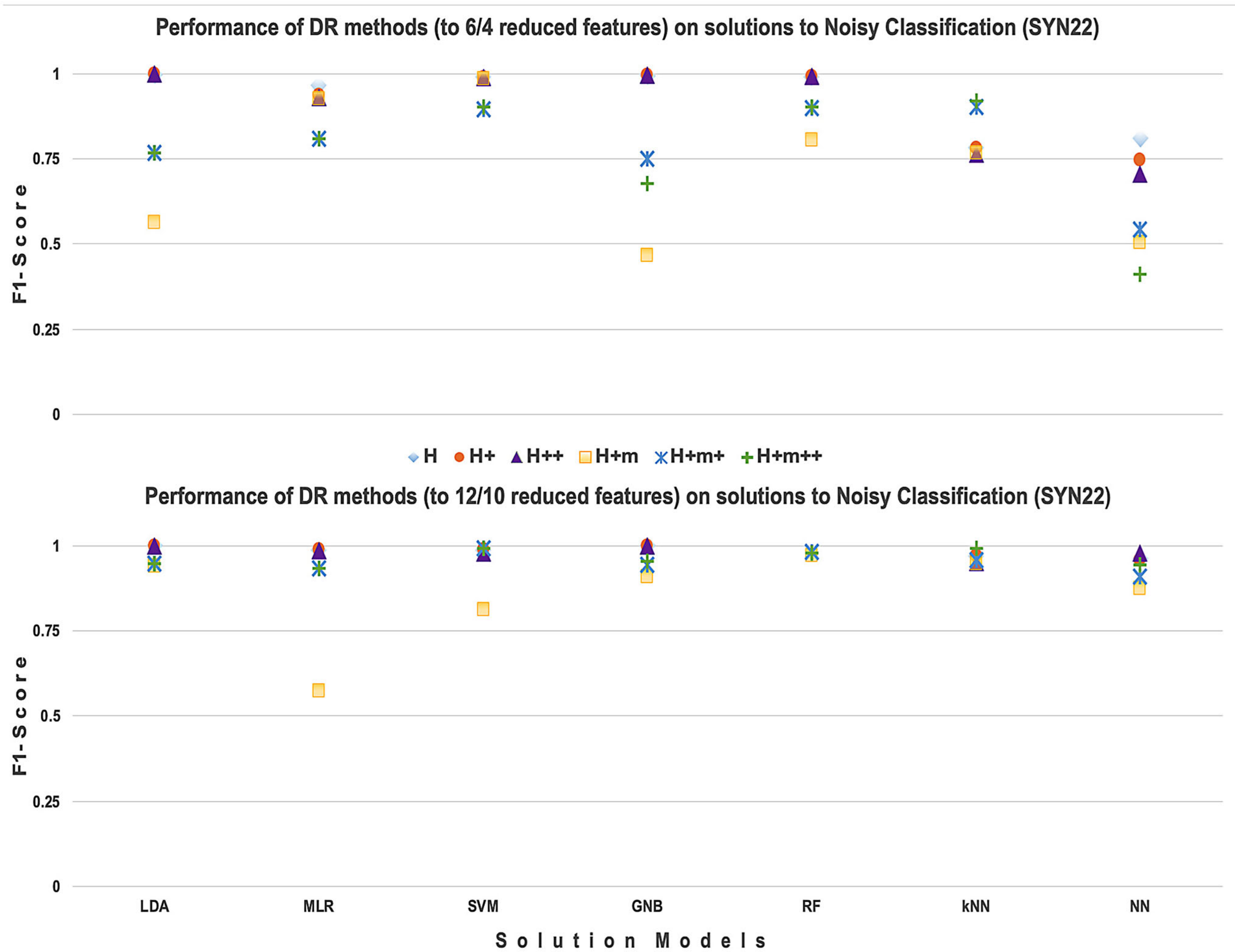

Fig. 6 Overall Performance of all DR methods based on conditional entropy for Noisy Classification based on models trained on 6 features (top) and 12 Features (bottom) with synthetic datasets, SYN22. X-axis: solutions (LDA, MLR, SVM, RF, GNB, kNN, NN.) Y-axis: F1-scores. The eight DR methods are as shown in Fig. 3

ning times of extracting the features increase exponentially with the dimensionality of the data. Third, importantly, their success can be easily *explained* because of the information-theoretic foundation behind them. Thus, the applications produced from these solutions may meet more acceptability than those produced by other methods since they can be easily rationalized.

Finally, these results leave some interesting questions unanswered. First, one is to make a systematic comparison on a common data set to other DR methods using related concepts, such as mutual information (MI) that likewise produce interesting results [4,46,51,54]. Second, in some sense, information-theoretic methods uncover a kind of knowledge, although very generically. Another intriguing question is thus whether domain knowledge can be more specifically used to reduce dimensionality in a data set for more effective solutions. Third, DNA methods offer the potential to apply the massive parallelism of DNA computing to process huge data sets *in vitro*. This would require developing more sophisticated methods to encode arbitrary (abiotic) digital data into DNA sequences.

Acknowledgements The use of HPC at the U of Memphis for processing datasets and training models is gratefully acknowledged. We are also grateful to the reviewers for valuable comments that resulted in substantial improvements to the quality and presentation of this work.

Funding Not applicable.

Data availability All data for the MC and BT problem are publicly available. The synthetic dataset used in the Noisy classification problem is available in the supplementary materials.

Code Availability The software used to run these applications is a part of the *sklearn package*, or is publicly available at *bmc.memphis.edu/DSAx/*, where 3D plots that allow full comparison of the results on all models and data sets can also be found.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Appendix

Table 5 Performance of DR methods to 6 features across various solution methods of Malware Classification

DR methods	The no. of reduced features	Solution method	F1 Score
H	6	LDA	0.3579
		MLR	0.6433
		SVM	0.7618
		GNB	0.3189
		RF	0.9746
		kNN	0.9593
		NN	0.6949
H+	6	LDA	0.7087
		MLR	0.8230
		SVM	0.8398
		GNB	0.1072
		RF	0.9883
		kNN	0.9125
		NN	0.7447
H++	6	LDA	0.8031
		MLR	0.8389
		SVM	0.9623
		GNB	0.8007
		RF	0.9850
		kNN	0.9722
		NN	0.7752
H+m	6	LDA	0.7797
		MLR	0.8075
		SVM	0.9310
		GNB	0.8026
		RF	0.9641
		kNN	0.9448
		NN	0.7051
H+m+	6	LDA	0.4540
		MLR	0.5933
		SVM	0.7538
		GNB	0.4314
		RF	0.9405
		kNN	0.8897
		NN	0.6939
H+m++	4	LDA	0.6097
		MLR	0.6779
		SVM	0.7620
		GNB	0.6537
		RF	0.9215
		kNN	0.8793
		NN	0.2851
3mE4b	6	LDA	0.9801
		MLR	0.7187
		SVM	0.8630
		GNB	0.9043
		RF	0.9885
		kNN	0.9411
		NN	0.5792

Table 6 Performance of DR methods to 12 features across various solution methods of Malware Classification

DR methods	The no. of reduced features	Solution method	F1 Score
H	12	LDA	0.4624
		MLR	0.8398
		SVM	0.8152
		GNB	0.5871
		RF	0.9869
		kNN	0.9690
		NN	0.8397
H+	12	LDA	0.7946
		MLR	0.9217
		SVM	0.9223
		GNB	0.6151
		RF	0.9895
		kNN	0.9323
		NN	0.9584
H++	12	LDA	0.8520
		MLR	0.9060
		SVM	0.9745
		GNB	0.8215
		RF	0.9883
		kNN	0.9639
		NN	0.8317
H+m	12	LDA	0.8624
		MLR	0.8950
		SVM	0.9713
		GNB	0.9096
		RF	0.9717
		kNN	0.9460
		NN	0.8227
H+m+	12	LDA	0.5500
		MLR	0.8066
		SVM	0.8921
		GNB	0.4445
		RF	0.9763

Table 6 continued

DR methods	The no. of reduced features	Solution method	F1 Score
		kNN	0.7467
		NN	0.8168
H+m++	12	LDA	0.6995
		MLR	0.7763
		SVM	0.8404
		GNB	0.7213
		RF	0.9498
		kNN	0.9210
		NN	0.7618
8mP10-4	10	LDA	0.7900
		MLR	0.5584
		SVM	0.6976
		GNB	0.5540
		RF	0.7327
		kNN	0.6986
		NN	0.4391
Related work	–	Kaggle Winner [58]	0.9935
		SNNMAC [63]	0.9919
		MalNet [61]	0.9855
		Hanqi Zhang [64]	0.9138
Common machine learning models [63]		RF	0.8338
		Xgboost	0.7802
		Naive Bayes	0.7013
		MLR	0.6934
		SVM	0.3772

Table 7 Performance of DR methods to 6 features across various solution methods of BioTaxonomy

DR methods	The no. of reduced features	Solution method	F1 Score
H	6	LDA	0.9041
		MLR	0.9427
		SVM	0.7975
		GNB	0.8613
		RF	0.9520
		kNN	0.9150
		NN	0.2796
H+	6	LDA	0.8854
		MLR	0.9281
		SVM	0.7737
		GNB	0.9118
		RF	0.9409
		kNN	0.9019
		NN	0.2831
H++	6	LDA	0.944
		MLR	0.9391
		SVM	0.8528
		GNB	0.9148
		RF	0.9683
		kNN	0.9591
		NN	0.2834
H+m	6	LDA	0.8182
		MLR	0.8973
		SVM	0.8181
		GNB	0.9032
		RF	0.9508
		kNN	0.9463
		NN	0.2976
H+m+	6	LDA	0.9149
		MLR	0.9184
		SVM	0.8544
		GNB	0.8884
		RF	0.9371
		kNN	0.9208
		NN	0.2674
H+m++	4	LDA	0.9414
		MLR	0.9369
		SVM	0.8492
		GNB	0.9238
		RF	0.9654
		kNN	0.9617
		NN	0.2851
3mE4b	6	LDA	0.8831
		MLR	0.9326
		SVM	0.6705
		GNB	0.8484
		RF	0.9589
		kNN	0.9182
		NN	0.6690

Table 8 Performance of DR methods to 12 features across various solution methods of BioTaxonomy

DR methods	The no. of reduced features	Solution method	F1 Score
H	12	LDA	0.972
		MLR	0.9404
		SVM	0.8208
		GNB	0.922
		RF	0.9602
		kNN	0.9182
		NN	0.7815
H+	12	LDA	0.9465
		MLR	0.9415
		SVM	0.8246
		GNB	0.9390
		RF	0.9567
		kNN	0.9398
		NN	0.6451
H++	12	LDA	0.9804
		MLR	0.9495
		SVM	0.8829
		GNB	0.888
		RF	0.9557
		kNN	0.9481
		NN	0.8834
H+m	12	LDA	0.92
		MLR	0.9001
		SVM	0.8795
		GNB	0.9268
		RF	0.9619
		kNN	0.9558
		NN	0.8267
H+m+	12	LDA	0.9643
		MLR	0.9357
		SVM	0.8846
		GNB	0.9270
		RF	0.9621
		kNN	0.9522
		NN	0.8168
H+m++	12	LDA	0.9675
		MLR	0.9491
		SVM	0.8808
		GNB	0.9188
		RF	0.9727
		kNN	0.9548
		NN	0.7619

Table 8 continued

DR methods	The no. of reduced features	Solution method	F1 Score
8mP10-4	10	LDA	0.4316
		MLR	0.4914
		SVM	0.4544
		GNB	0.4161
		RF	0.4949
		kNN	0.4490
		NN	0.4279
Related work [62]		SVM	0.9688
		Naive Bayes	0.8558
		RF	0.9690
		kNN	0.9036
		NN	0.9345
		DeepBarcoding	0.9763

Table 9 Performance of DR methods to 6 features across various solution methods of Noisy Classification (SYN13)

DR methods	The no. of reduced features	Solution method	F1 Score
H+m	6	LDA	0.3992
		MLR	0.9805
		SVM	0.9912
		GNB	0.2601
		RF	0.7567
		kNN	0.6979
		NN	0.2801
H++	6	LDA	1
		MLR	0.9711
		SVM	0.9935
		GNB	0.7757
		RF	0.9792
		kNN	0.9810
		NN	0.8238
H	6	LDA	1
		MLR	0.9805
		SVM	0.9912
		GNB	0.8598
		RF	0.9806
		kNN	0.9840
		NN	0.9153
H+	6	LDA	1
		MLR	0.9811
		SVM	0.9922
		GNB	0.8598
		RF	0.9809
		kNN	0.9840
		NN	0.9232

Table 9 continued

DR methods	The no. of reduced features	Solution method	F1 Score
H+m++	6	LDA	0.6936
		MLR	0.779
		SVM	0.9900
		GNB	0.5151
		RF	0.9879
		kNN	0.99
		NN	0.7237
H+m+	4	LDA	0.6937
		MLR	0.7794
		SVM	0.9900
		GNB	0.5151
		RF	0.9879
		kNN	0.9900
		NN	0.6939

Table 10 Performance of DR methods to 12 features across various solution methods of Noisy Classification (SYN13)

DR methods	The no. of reduced features	Solution method	F1 Score
H+m	12	LDA	0.9901
		MLR	0.4462
		SVM	0.7097
		GNB	0.9353
		RF	0.9931
		kNN	0.9860
		NN	0.9191
H++	12	LDA	1
		MLR	0.9902
		SVM	0.9924
		GNB	0.9369
		RF	0.9864
		kNN	0.9709
		NN	0.9605
H	12	LDA	1
		MLR	0.9828
		SVM	0.9946
		GNB	0.9358
		RF	0.9865
		kNN	0.9709
		NN	0.9826

Table 10 continued

DR methods	The no. of reduced features	Solution method	F1 Score
H+	12	LDA	1
		MLR	0.9830
		SVM	0.9946
		GNB	0.9358
		RF	0.9867
		kNN	0.9709
		NN	0.9807
H+m++	12	LDA	1
		MLR	0.9819
		SVM	0.9951
		GNB	0.9781
		RF	0.9905
		kNN	0.9761
		NN	0.9781
H+m+	12	LDA	0.8307
		MLR	0.9798
		SVM	0.9885
		GNB	0.9464
		RF	0.9891
		kNN	0.9712
		NN	0.9249
Related Work	–	[38]	0.7970
		[1]	0.9600

Table 11 Performance of DR methods to 6 features across various solution methods of Noisy Classification (SYN22)

DR methods	The no. of reduced features	Solution method	F1 Score
H	6	LDA	0.5635
		MLR	0.9246
		SVM	0.9865
		GNB	0.4654
		RF	0.8063
		kNN	0.7657
		NN	0.5029
H+	6	LDA	0.9975
		MLR	0.9292
		SVM	0.9875
		GNB	0.9940
		RF	0.9910
		kNN	0.7646
		NN	0.7033
H++	6	LDA	0.9973
		MLR	0.9651
		SVM	0.9909
		GNB	0.9950
		RF	0.9910
		kNN	0.7821
		NN	0.8111
H+m	6	LDA	0.9973
		MLR	0.9354
		SVM	0.9900
		GNB	0.9950
		RF	0.9910
		kNN	0.7821
		NN	0.7447
H+m+	6	LDA	0.7667
		MLR	0.8096
		SVM	0.9023
		GNB	0.6773
		RF	0.9023
		kNN	0.9188
		NN	0.41208
H+m++	4	LDA	0.7668
		MLR	0.8096
		SVM	0.8963
		GNB	0.7487
		RF	0.8981
		kNN	0.9036
		NN	0.5427

Table 12 Performance of DR methods to 12 features across various solution methods of Noisy Classification (SYN22)

DR methods	The no. of reduced features	Solution method	F1 Score
H	12	LDA	0.9402
		MLR	0.5728
		SVM	0.8104
		GNB	0.9096
		RF	0.9717
		kNN	0.946
		NN	0.8726
H+	12	LDA	1
		MLR	0.9849
		SVM	0.9783
		GNB	1
		RF	9895
		kNN	0.9511
		NN	0.9779
H++	12	LDA	1
		MLR	0.9875
		SVM	0.9877
		GNB	1
		RF	9917
		kNN	0.9699
		NN	0.9609
H+m	12	LDA	1
		MLR	0.9876
		SVM	0.9878
		GNB	1
		RF	9917
		kNN	0.9699
		NN	0.9584
H+m+	12	LDA	0.9481
		MLR	0.9345
		SVM	0.9906
		GNB	0.9547
		RF	0.9794
		kNN	0.9910
		NN	0.9417
H+m++	12	LDA	0.9484
		MLR	0.9345
		SVM	0.9902
		GNB	0.9425
		RF	0.9800
		kNN	0.9578
		NN	0.9082
Related work	–	[38]	0.7970
		[1]	0.9600

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